Project Initialization and Planning Phase

Date	29 October 2026
Team ID	N/A - individual project; no Team ID provided
Project Title	Electric Vehicle Market Performance & Insights
Maximum Marks	3 Marks

Project Proposal (Proposed Solution) template

This project proposal outlines a solution to address a specific problem. With a clear objective, defined scope, and a concise problem statement, the proposed solution details the approach, key features, and resource requirements, including hardware, software, and personnel.

EV Market Intelligence is a descriptive analytics prototype for exploring a synthetic electric-vehicle dataset across categories, manufacturers, geography, model year and battery technology.	
Objective	Provide an accessible, interactive view of recorded EV attributes using repeatable preparation, SQL analysis, Tableau visualizations and web embedding.
Scope	Cleaned training data, SQLite database/queries, nine Tableau chart worksheets, one dashboard with filters, three-scene Story, Flask embed website and documentation. No predictive modeling or real-world market validation.
A broad dataset with many attributes is difficult to compare manually across EV categories, manufacturers, regions and battery technologies.	
Description	The source has 200,000 rows before cleaning, 37 columns and one exact duplicate. Five numeric columns have missing values requiring documented treatment.
Impact	Makes the synthetic dataset easier to summarize, compare and explain; provides an end-to-end example of data preparation, SQL, visual analytics and deployment.
Clean data, load a SQLite database, perform SQL operations, create Tableau charts/dashboard/Story and embed published views in Flask.	
Approach	Python/pandas preparation; SQLite schema and SQL query pack; Tableau Public connected to cleaned CSV; Flask + Tableau Embedding API; Render hosting.
Key Features	Four dashboard filters; treemap, packed bubbles, bar chart, line chart, pie chart, map, heat map, scatter plot and dot plot; three narrative scenes; KPI/API endpoints.

Resource Requirements

Resource Type	Description	Specification/Allocation
User workstation; exact specifications not recorded.		
Computing Resources	Desktop/laptop for Tableau, Python and browser	No GPU requirement; CPU model/core count not recorded.
Memory	Local data processing and Tableau authoring	Installed RAM not recorded.

Storage	CSV, SQLite DB, workbook, templates and app source	CSV ~57 MB and SQLite DB ~73 MB in supplied kit; exact OS disk capacity not recorded.
Windows; Tableau Public Desktop; GitHub Desktop; Python/Flask; browser.		
Frameworks	Web application framework	Flask; Gunicorn for production hosting as configured.
Libraries	Data and web dependencies	pandas for preparation in the supplied project assets; Flask and python-dotenv in requirements; Tableau Embedding API v3 in page.
Development Environment	Source and authoring tools	Antigravity, Tableau Public Desktop, GitHub Desktop; Git repo initialized locally.
Kaggle dataset link supplied through SkillWallet; exact URL to be copied from portal. Synthetic/procedurally generated training records.		
Data Size	Cleaned analytical dataset	199,999 records, 37 columns; SQLite table ev_market_clean.